THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 81	First term + (12-1) x 7 = 4+11x7
2	**(b)** 240	The two angles between the hands add up to 360°.
3	**(a)** 1234	End with the smallest even digit, then arrange the rest in increasing order.
4	**(d)** 100	Add the two biggest numbers.
5	**(d)** 10	$48 = 2^4 \times 3$; use $(e_1+1)(e_2+1)\ldots$
6	**(c)** 126	Find the breadth first.
7	**(a)** $3\frac{1}{6}$	Convert to improper fractions or borrow.
8	**(a)** 2	A point is inside if its distance from O is less than the radius.
9	**(c)** 18	Lines of symmetry = $360 \div$ angle.
10	**(a)** -13	Add step by step.
11	**(d)** 18	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
12	**(c)** 120	6 degrees per minute, or 30° per number on the dial.
13	**(c)** 90	First digit 1-9 and middle digit 0-9, the last digit is fixed: 9×10.
14	**(d)** 75	Find the difference in symbols then multiply.
15	**(b)** 12	Highest common factor.
16	**(a)** 144	Area becomes 4 times.
17	**(b)** 1	Use the LCM of all denominators.
18	**(b)** 180	Triangle angle sum.
19	**(d)** 300	Multiply.
20	**(a)** 8	Add the opposites.
21	**(a)** 156	Sum of first n even numbers is $n \times (n+1)$.
22	**(d)** 5	30° per hour.
23	**(b)** 29992	Fill the middle digits with 9.
24	**(b)** 165	Total height times scale.
25	**(b)** 36	Lowest common multiple.
26	**(c)** 4800	Area $\times$ rate.
27	**(a)** 60	Divide by 3, multiply by 5.
28	**(a)** 130	Linear pair.
29	**(d)** A square	Only the square among these has 4.
30	**(b)** 0	Opposites cancel.
31	**(a)** 210	Add 20.
32	**(b)** 160	Full turn is 360°.
33	**(c)** 4	Count each step.
34	**(a)** $\frac{6}{29}$	Fraction $= \frac{30}{145}$ (simplified)
35	**(b)** 1	Digit sum must be a multiple of 3.
36	**(c)** 9	Perimeter of square $= 4\times$ side.
37	**(b)** $\frac{5}{8}$	Fraction left $= 1-$ fraction spent.
38	**(a)** 5	Outer radius minus inner radius.
39	**(d)** 60	The square fits after every multiple of 90°.
40	**(c)** 17	Highest minus lowest.
41	**(d)** 21	Sticks $= 3 + 2 \times (n-1) = 2n+1$.

42	**(d)** 27	A bisector divides the angle into two equal parts.
43	**(c)** 1441	Keep the first half and mirror it; if that is not bigger, increase the middle.
44	**(b)** 65	Add the symbols, then multiply by the value of one symbol.
45	**(c)** 994	Divide 999 by the number.
46	**(d)** 100	Side of square $=$ perimeter $\div 4$.
47	**(c)** $\frac{3}{8}$	Take the average of the two fractions.
48	**(c)** 8	Side $=$ perimeter $\div 4$.
49	**(d)** 10	Add.
50	**(b)** -30	Subtract the descent.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 625	Consecutive triangular numbers add to a square.
2	**(a)** 270	Add the two turns in degrees.
3	**(a)** 465	Pair the numbers: first with last.
4	**(d)** 0	Count.
5	**(c)** 9	Count them carefully.
6	**(d)** 256	Inner side $= a - 2 \times 2$.
7	**(c)** 17	Whole times denominator plus numerator.
8	**(b)** 24	4 diameters.
9	**(d)** 5	Count the letters with a fold line.
10	**(c)** -7 and 12	Check each pair.
11	**(d)** 128	Each term is double the one before it.
12	**(c)** right	Find the angle first.
13	**(a)** 6174	Largest - smallest.
14	**(d)** 10	Convert the increase into cm.
15	**(d)** 17	$210 = 2 \times 3 \times 5 \times 7$
16	**(c)** 9	Height $= 2\times$ area $\div$ base.
17	**(d)** $\frac{3}{7}$	Divide numerator and denominator by their HCF.
18	**(a)** 7	Half of AB.
19	**(d)** 45	$360\div$ spokes.
20	**(c)** 4	Count the positive integers up to b-1.
21	**(c)** 93	First term + (15-1) x 6 = 9+14x6
22	**(a)** 75	Straight angle is 180°.
23	**(c)** 9100	Put 9s from the left until the sum is used up.
24	**(d)** 135	Add the two biggest numbers.
25	**(a)** 8	$54 = 2 \times 3^3$; use $(e_1+1)(e_2+1)\ldots$
26	**(c)** 54	Half of the rectangle's area.
27	**(d)** 21	Divide, then multiply.
28	**(d)** 3	A point is inside if its distance from O is less than the radius.
29	**(b)** 20	Lines of symmetry = $360\div$ angle.
30	**(b)** -3	Add step by step.
31	**(d)** 25	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
32	**(d)** 210	The two angles between the hands add up to 360°.
33	**(d)** 2576	End with the smallest even digit, then arrange the rest in increasing order.
34	**(b)** 125	Find the difference in symbols then multiply.
35	**(a)** 15	Highest common factor.
36	**(a)** 154	Area $= l \times b$.
37	**(c)** $2\frac{11}{12}$	Convert to improper fractions or borrow.
38	**(b)** 34	Opposite sides are equal.
39	**(d)** 225	Multiply.
40	**(c)** -10	Add the opposites.
41	**(b)** 240	Sum of first n even numbers is $n \times (n+1)$.

42	**(d)** 210	6 degrees per minute, or 30° per number on the dial.
43	**(b)** 49994	Fill the middle digits with 9.
44	**(b)** 275	Total height times scale.
45	**(a)** 40	Lowest common multiple.
46	**(a)** 275	Find the breadth first.
47	**(c)** $\frac{3}{4}$	Use the LCM of all denominators.
48	**(b)** 110	Linear pair.
49	**(b)** hexagon	6 lines mean 6 sides.
50	**(c)** 0	Opposites cancel.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 300	Add 24.
2	**(c)** 7	30° per hour.
3	**(c)** 7	Count each step.
4	**(c)** $\frac{11}{29}$	Fraction $= \frac{55}{145}$ (simplified)
5	**(a)** 2	Digit sum must be a multiple of 3.
6	**(a)** 324	Area becomes 4 times.
7	**(d)** 63	Divide by 2, multiply by 7.
8	**(d)** 6	Outer radius minus inner radius.
9	**(b)** 40	The square fits after every multiple of 90°.
10	**(d)** 19	Highest minus lowest.
11	**(d)** 31	Sticks $= 3 + 2 \times (n-1) = 2n+1$.
12	**(a)** 120	Full turn is 360°.
13	**(d)** 4884	Keep the first half and mirror it; if that is not bigger, increase the middle.
14	**(b)** 144	Add the symbols, then multiply by the value of one symbol.
15	**(d)** 988	Divide 999 by the number.
16	**(b)** 9000	Area $\times$ rate.
17	**(b)** $\frac{3}{5}$	Fraction left $= 1-$ fraction spent.
18	**(a)** 11	Side $=$ perimeter $\div 4$.
19	**(a)** 8	Add.
20	**(a)** -15	Subtract the descent.
21	**(d)** 900	Consecutive triangular numbers add to a square.
22	**(b)** 38	A bisector divides the angle into two equal parts.
23	**(a)** 1035	Pair the numbers: first with last.
24	**(a)** 5	Count.
25	**(c)** 9	Count them carefully.
26	**(a)** 12	Perimeter of square $= 4\times$ side.
27	**(d)** $\frac{1}{2}$	Take the average of the two fractions.
28	**(a)** 32	4 diameters.
29	**(a)** 2	Count the letters with a fold line.
30	**(c)** -9 and 4	Check each pair.
31	**(b)** 256	Each term is double the one before it.
32	**(a)** 210	Add the two turns in degrees.
33	**(d)** 8532	Largest - smallest.
34	**(d)** 9	Convert the increase into cm.
35	**(c)** 21	$330 = 2 \times 3 \times 5 \times 11$
36	**(a)** 196	Side of square $=$ perimeter $\div 4$.
37	**(c)** 35	Whole times denominator plus numerator.
38	**(b)** 9	Half of AB.
39	**(a)** 40	$360\div$ spokes.
40	**(b)** 7	Count the positive integers up to b-1.
41	**(c)** 93	First term + (11-1) x 8 = 13+10x8
42	**(c)** straight	Find the angle first.

43	**(b)** 9600	Put 9s from the left until the sum is used up.
44	**(a)** 80	Add the two biggest numbers.
45	**(a)** 12	$72 = 2^3 \times 3^2$; use $(e_1+1)(e_2+1)\ldots$
46	**(d)** 400	Inner side $= a - 2 \times 2$.
47	**(c)** $\frac{5}{8}$	Divide numerator and denominator by their HCF.
48	**(d)** 2	A point is inside if its distance from O is less than the radius.
49	**(d)** 24	Lines of symmetry = $360 \div$ angle.
50	**(b)** -3	Add step by step.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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